

Ans 9.4.1 $\langle n=1 | \frac{1}{x^2} | n=1 \rangle$

$$\psi_1(x) = \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} \exp\left(-\frac{m\omega x^2}{2\hbar}\right) \sqrt{\frac{m\omega}{\hbar}} x$$

$$\langle \frac{1}{x^2} \rangle = \int_{-\infty}^{\infty} \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} \exp\left(-\frac{m\omega x^2}{2\hbar}\right) \left(\frac{m\omega}{\hbar} \right)^{1/4} \frac{1}{x^2} x^2 dx$$

$$= 4 \left(\frac{m\omega}{4\pi \hbar} \right)^{1/2} \left(\frac{m\omega}{\hbar} \right)^{1/4} \int_{-\infty}^{\infty} \exp\left(-\frac{m\omega x^2}{2\hbar}\right) dx$$

$$= 4 \left(\frac{m\omega}{4\pi \hbar} \right)^{1/2} \left(\frac{m\omega}{\hbar} \right)^{1/4} 2 \int_0^{\infty} \exp\left(-\frac{m\omega x^2}{2\hbar}\right) dx = 4 \left(\frac{m\omega}{4\pi \hbar} \right)^{1/2} \left(\frac{m\omega}{\hbar} \right)^{1/4} \left(\frac{\pi \hbar}{m\omega} \right)^{1/2}$$

$$= 2 \left(\frac{m\omega}{\hbar} \right)$$

(7)

$$\langle n=1 | x^2 | n=1 \rangle = \int_{-\infty}^{\infty} \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} \exp\left(-\frac{m\omega x^2}{2\hbar}\right) \left(\frac{m\omega}{\hbar} \right)^{1/4} x^4 dx$$

$$= 4 \left(\frac{m\omega}{4\pi \hbar} \right)^{1/2} \left(\frac{m\omega}{\hbar} \right)^{1/4} \int_{-\infty}^{\infty} x^4 \exp\left(-\frac{m\omega x^2}{2\hbar}\right) dx$$

$$= 4 \left(\frac{m\omega}{4\pi \hbar} \right)^{1/2} \left(\frac{m\omega}{\hbar} \right)^{1/4} \sqrt{\frac{\hbar \pi}{m\omega}} \frac{(2\omega-1)!!}{(2\frac{m\omega}{\hbar})^2} = \frac{4}{2} \left(\frac{m\omega}{\pi \hbar} \right)^{1/2} \left(\frac{m\omega}{\hbar} \right)^{1/4} \left(\frac{\hbar \pi}{m\omega} \right)^{1/2} \frac{3}{4 \left(\frac{m\omega}{\hbar} \right)^2}$$

$$= \frac{3 \hbar}{2 m\omega}$$

$$\therefore \frac{1}{\langle n=1 | x^2 | n=1 \rangle} = \frac{2 m\omega}{3 \hbar}$$

Thus,

$$\langle \frac{1}{x^2} \rangle \approx \frac{1}{\langle x^2 \rangle} \approx \frac{m\omega}{\hbar}$$

(2)

Ans 9.4.2. $\Psi = \frac{1}{(\pi a_0^3)^{1/2}} e^{-r/a_0}$

$$\langle \Psi | \Psi \rangle = 4\pi \int_0^\infty \frac{1}{(\pi a_0^3)} e^{-2r/a_0} r^2 dr = \frac{4}{a_0^3} \int_0^\infty e^{-2r/a_0} r^2 dr$$

$$= \frac{4}{a_0^3} \frac{2}{(2/a_0)^3} = \frac{8}{8} \frac{a_0^3}{a_0^3} = \boxed{1}$$

$x = \left(\frac{2}{a_0}\right)$
 $n = 2$

(2) $(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2$

We have spherical wavefunction. So $\langle x \rangle = \langle y \rangle = \langle z \rangle = 0$

$$\therefore (\Delta x)^2 = \langle x^2 \rangle$$

$$\langle x^2 \rangle + \langle y^2 \rangle + \langle z^2 \rangle = \langle r^2 \rangle$$

($\because$ spherical, symmetric w.r.t x, y & z)

(8) $\therefore \langle x^2 \rangle = \frac{1}{3} \langle r^2 \rangle$

$$= \frac{1}{3} 4\pi \int_0^\infty \frac{1}{\pi a_0^3} e^{-2r/a_0} r^4 dr$$

$$= \frac{4}{3 a_0^3} \int_0^\infty e^{-2r/a_0} r^4 dr = \frac{4}{3 a_0^3} \frac{4!}{(2/a_0)^5} = \frac{4}{3} a_0^2 \frac{4 \cdot 3 \cdot 2 \cdot 1}{32}$$

$$\therefore (\Delta x)^2 = a_0^2 \Rightarrow \boxed{(\Delta x) = a_0 = \frac{\hbar^2}{me^2}}$$

(3) $\langle \frac{1}{r} \rangle \xrightarrow{\text{in coordinate}} 4\pi \int_0^\infty \frac{1}{(\pi a_0^3)} e^{-2r/a_0} r dr$

$$= \frac{4}{a_0^3} \int_0^\infty r e^{-2r/a_0} dr = \frac{4}{a_0^3} \frac{1}{(2/a_0)^2} = \frac{1}{a_0} = \frac{me^2}{\hbar^2}$$

$$\langle r \rangle = 4\pi \int_0^\infty \frac{1}{\pi a_0^3} e^{-2r/a_0} r^3 dr = \left(\frac{4\pi}{\pi a_0^3} \right) \left(\frac{3 a_0^4}{8} \right) = \frac{3}{2} a_0 = \frac{3}{2} \frac{\hbar^2}{me^2}$$

$$\therefore \frac{1}{\langle r \rangle} = \frac{2}{3} \frac{me^2}{\hbar^2} \therefore \langle \frac{1}{r} \rangle = \frac{1}{\langle r \rangle} = \frac{me^2}{\hbar^2}$$

Ans 3.4.3. $H = \frac{p^2}{2m} - \frac{e^2}{(R^2)^{1/2}}$; $\langle H \rangle = \frac{\langle p^2 \rangle}{2m} - \frac{e^2}{\langle R^2 \rangle^{1/2}}$

$$(\Delta p)^2 = \langle p^2 \rangle - \langle p \rangle^2$$

$$(\Delta R)^2 = \langle R^2 \rangle - \langle R \rangle^2$$

We are interested in ground state

$\Rightarrow \langle H \rangle$ minimum. So, work with

states in which $\langle p \rangle = \langle R \rangle = 0$

⑦ $\therefore \langle H \rangle \approx \frac{(\Delta p)^2}{2m} - \frac{e^2}{[\Delta R]^2]^{1/2}}$ $(\Delta p)(\Delta R) \geq \frac{\hbar}{2}$

$$\therefore \langle H \rangle \geq \frac{\hbar^2}{8(\Delta R)^2 m} - \frac{e^2}{(\Delta R)}$$

$$\therefore (\Delta p) \geq \frac{\hbar}{2(\Delta R)}$$

Differentiating, $\frac{\hbar^2(-2)}{8m(\Delta R)^3} + \frac{e^2}{(\Delta R)^2} = 0$

$$\therefore \frac{e^2}{(\Delta R)^2} = \frac{2\hbar^2}{8m(\Delta R)^3} \quad \therefore e^2 = \frac{\hbar^2}{4m(\Delta R)} \quad \therefore (\Delta R) = \frac{\hbar^2}{4me^2}$$

$$\therefore \langle H \rangle \geq \frac{\hbar^2}{8m \left(\frac{\hbar^2}{4me^2} \right)^2} - \frac{e^2 (4me^2)}{\hbar^2}$$

$$\geq \frac{2\hbar^2 4m^2 e^4}{4m \hbar^4} - \frac{4e^4 m}{\hbar^2}$$

$$\geq \frac{2e^4 m}{\hbar^2} - \frac{4e^4 m}{\hbar^2} \quad \therefore \boxed{\langle H \rangle \geq -\frac{2me^4}{\hbar^2}}$$

$\therefore$ The ground state energy is of the order of $-\frac{2me^4}{\hbar^2}$

Ans 9.4.4

(4)

$$(\Delta Q)^2 (\Delta P)^2 \geq \frac{1}{4} \langle \Psi | [\hat{Q}, \hat{P}]_+ | \Psi \rangle^2 + \frac{1}{4} \langle \Psi | \hat{F} | \Psi \rangle^2$$

where, $\hat{Q} = \hat{Q} - \bar{Q}$, $\hat{P} = \hat{P} - \bar{P}$; $[\hat{Q}, \hat{P}] = [\hat{Q}, \hat{P}] = i\hat{F}$

$$[T, X] = \frac{1}{2m} [P^2, X] = P [P, X] + [P, X] P = P(-i\hbar) - i\hbar P$$

$$= -2i\hbar P$$

$\therefore$ Here $\hat{F} = -2\hbar P$.

$[\hat{Q}, \hat{P}]_+$ is Hermitian $\therefore \langle [\hat{Q}, \hat{P}]_+ \rangle^2 \geq 0$

(9) $\therefore (\Delta T)^2 (\Delta X)^2 \geq \frac{1}{4} \langle \hat{P} \rangle^2 \hbar^2$ $(\Delta T)(\Delta X) \geq \frac{\hbar |P|}{2m}$

This relation is not paravo because it involves the particle momentum, $\langle p \rangle$ & therefore is not universal.

Also for bound states $\langle p \rangle = 0$. This relation just gives $(\Delta T)(\Delta X) \geq 0$, which is trivial information.